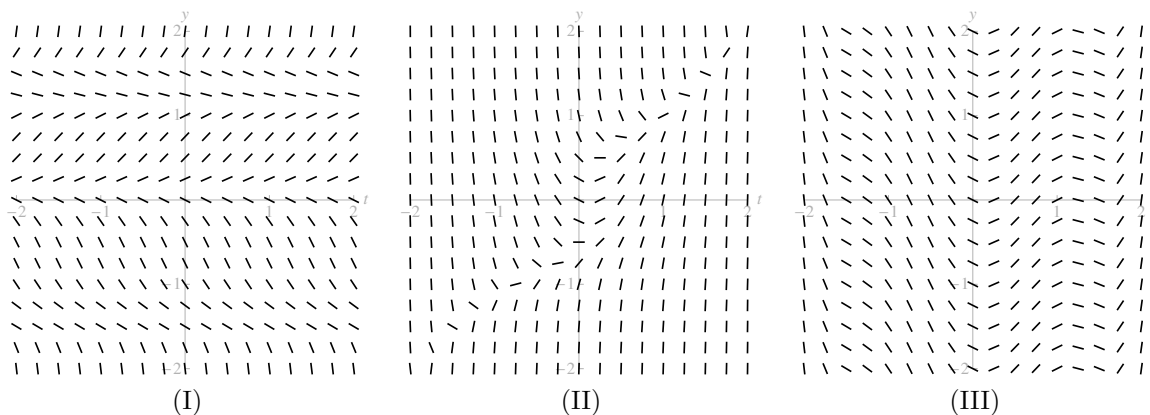


04 - Autonomous First-Order ODEs

1. **Daily Problem.** Here are three direction fields.



For each of the following differential equations, find the picture above which shows its direction field.

(a) $\frac{dy}{dt} = t^5 - 4t^3 + 4t - \frac{1}{2}.$

(b) $\frac{dy}{dt} = y^5 - 4y^3 + 4y - \frac{1}{2}.$

(c) For each differential equation, can you also draw a few representative solutions? Are some solutions very similar to others, and if so, how? Do some of the solutions cross?

(d) Today we'll focus on differential equations such as (b). Can you find, using its direction field, solutions of (b) that remain constant as t changes? Draw them on the direction field.

2. Let $P(t)$ represent the population of a species of fish in a lake. Here's a simple model for $P(t)$, assuming L is the maximum number of fish the lake can sustain:

$$\frac{dP}{dt} = kP(L - P) \quad \text{is called the } \underline{\text{logistic equation}}.$$

(a) Give two quantities that dP/dt is proportional to, and make sense of that in the given context.

(b) Sketch a direction field for this differential equation. Try to do as little work as possible.

3. Use qualitative analysis to sketch a family of representative solutions for each differential equation. Identify the equilibrium solutions of each differential equation, and say whether they are stable or unstable.

(a) $\frac{dy}{dt} = (y + 1)(4 - y).$

(b) $\frac{dy}{dt} = (y + 1)(y - 2)^2.$

4. IDE How could we describe all solutions to an autonomous differential equation using a one dimensional representation? (That's being even lazier, since our quantitative analysis, or lazy direction fields, from the previous problem is still a two dimensional representation!)

5. **LODE** A **phase line** is the standard one-dimensional diagram that depicts the qualitative behavior of solutions to an autonomous differential equation. Label the dots and add arrows to the figure below to represent **all** solutions to the differential equation in $\frac{dP}{dt} = \frac{P}{2} \left(1 - \frac{P}{5}\right) \left(\frac{P}{8} - 1\right)$.



6. **LODE** It can be difficult in practice to obtain an accurate number for an initial condition (say if y represents the number of fish in a lake — counting all the fish is impossible).
- For the ODE in #3(a), consider the initial conditions $y = 3.9$, $y = 4$, and $y = 4.1$. Give long-term predictions for the value of y using each of these initial conditions. Carry out a similar analysis if the initial condition is somewhere around -1 instead (assuming y can be negative).
 - For the ODE in #3(b), carry out a similar analysis if the initial condition is somewhere around 0 (notice $y = 0$ is not an equilibrium solution), and if it is somewhere around 2.
 - What word would you use to describe the equilibrium solution $y = 4$ for the ODE in #3(a), given your analysis above? What about the equilibrium solution $y = -1$ for the ODE in #3(a)? What about the equilibrium solution $y = 2$ for the ODE in #3(b)?
 - In all parts of this question, we gave you initial values of y , but we did not give an initial time. Did you assume $t = 0$? How does changing the initial time change your answers?
7. Consider again the differential equation $\frac{dy}{dt} = (y + 1)(4 - y)$ from #3(a). Explain in a few words how you expect the solutions to differ if we multiply the right-hand side by 5 and consider instead $\frac{dy}{dt} = 5(y + 1)(4 - y)$? What if we multiply the right-hand side by 0.5 and consider $\frac{dy}{dt} = 0.5(y + 1)(4 - y)$?
8. What does $f(3) = 5$ mean? What does $f(t) = 5$ mean? Why do you think this question is being asked of you today?
9. **LODE** A group of biologists are making predictions about the spotted owl population in a forest in the Canadian Rockies. The autonomous differential equation the scientists use to model the spotted owl population is $\frac{dP}{dt} = \frac{P}{2} \left(1 - \frac{P}{5}\right) \left(\frac{P}{8} - 1\right)$, where P is in hundreds of owls and t is in years. Sketch the phase line for this differential equation. Explain the similarities and differences between the behavior of solutions here vs in the case of the logistic equation of problem #2. How could that make sense given the particular context?